Hot-Swap eFuse (LTC4215) — Digital Telemetry Front-End

STM32F411CEU6 over I^2C

IKGB105

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1 Project Objective

`EFUSE_LTC4215` (the project this one extends) deliberately left the LTC4215's I^2C/SMBus monitoring interface unused – every ADR/SCL/SDA pin was wired no-connect, and the datasheet's onboard ADC (current, voltage, fault status) was flagged in that project's own *Known Limitations* as "real LTC4215 features left entirely untested here."

This project closes that gap: it adds a minimal STM32F411CEU6 system to the same board, wired to the LTC4215's I^2C bus, turning the eFuse from a purely analog hot-swap/protection circuit into one with digital telemetry read-out. It is the Phase 3 "digital power monitoring" item of the broader hardware roadmap (hot-swap → digital power/PMBus-style monitoring → PDN analysis), chosen specifically because the LTC4215's monitoring interface was already sitting there unused rather than as a new, unrelated MCU exercise.

Scope, same convention as every project in this series: **simulation + schematic + PCB layout only, not physically built**. A COMSOL thermal FEM was already carried out for the eFuse power stage in `EFUSE_LTC4215`; this project does not repeat it (Section 7).

Pending

Firmware (I^2C register read-out, compiled against the HAL but not flashed/tested on real hardware) and a PDN (power distribution network) impedance-vs-frequency check of the 3.3 V rail are both still open – see Section 8.

2 MCU Selection

STM32F411CEU6 (UFQFPN48), the same part used on the WeAct "Blackpill" board. Chosen over a smaller part (e.g. STM32F030) for two concrete reasons, not just familiarity:

- Direct USB 2.0 FS peripheral (OTG_FS) for USB-DFU flashing, matching a workflow already known from the real Blackpill, without needing an external SWD programmer.
- More compute/memory headroom than strictly required for I^2C register read-out, left available for a possible future command-parsing layer (closer to true PMBus-style interaction) without a second board respin.

The real WeAct Blackpill schematic (`docs/reference/`) was used as the primary layout reference, cross-checked and corrected in several places – documented as pitfalls in Section 4.4 rather than copied blindly.

3 Design

3.1 I²C integration with the LTC4215

The LTC4215's I²C-related pins, all left unconnected in the original eFuse design, are now wired per the datasheet (Rev E, "I²C Interface" and pin function sections):

- **SDAI (pin 8) and SDAO (pin 7) tied together**, forming the single bidirectional SDA line – the QFN package exposes them separately specifically to allow an optoisolator on SDAO alone; not used here, so they are simply shorted.
- **SCL (pin 9)** to the bus directly.
- **ADR0 (pin 12) and ADR2 (pin 14) tied to INTVCC**, **ADR1 (pin 13) left floating (No Connect)** – reproducing the datasheet's own worked example (Figure 5): this combination sets I²C address **19** (binary `1010011`), chosen for traceability to the datasheet rather than an arbitrary new value, since this is the only device on the bus and the exact address has no functional consequence.
- **GPIO (pin 18)** kept as in the original design (power-good flag); **-ALERT (pin 10)**, previously unused, now routed to the STM32 as an EXTI interrupt source.
- Pull-ups: 4.7 kΩ **on SCL and SDA**, 10 kΩ **on ALERT** (slower interrupt line, no need to match the bus speed resistors), all to **INTVCC** (≈3.1 V typ regulated, close enough to the STM32's 3.3 V logic levels that no level shifter is needed – confirmed from the datasheet's own electrical table, not assumed).

3.2 STM32 minimal system

- **Power**: three VDD pins (24/36/48) and VSS at pins 23/35/47/49 (the last one being the UFQFPN48's exposed pad – the default KiCad symbol represents it as a fourth, hidden VSS pin stacked on top of pin 23, electrically merged by position, not a separate net to manage; verified directly against the symbol library file, Section 4.4). VBAT tied directly to 3.3 V with its own 100 nF – no RTC battery backup is implemented, so the WeAct reference's BAT54C diode-OR network was simplified away.
- **VDDA/VREF+** filtered from 3.3 V through a 10 µH ferrite (L1) plus a bank of 100 nF decoupling, isolating the ADC/PLL analog supply from digital switching noise.
- **VCAP1 (pin 22)**: 2.2 µF, required for the F4's internal 1.2 V core regulator.
- **Clock**: 25 MHz crystal (Y2, 9 pF load, 3225 SMD package) with 12 pF loading caps on PH0/PH1 – no 32.768 kHz RTC crystal, intentionally dropped (Section 4.4).
- **Reset / Boot**: NRST with 100 nF only (the STM32 has an internal pull-up); BOOT0 with a 10 kΩ pull-down plus a push-button to 3.3 V to force USB-DFU entry; **BOOT1 (shared with PB2 on the F401/F410/F411/F412 value line, confirmed against RM0383)** likewise pulled down – both are genuinely required on this part, not a copy-paste artifact from the F1 series (Section 4.4).
- **USB-C**: D+/D- through 10 Ω series resistors to PA12/PA11; CC1 and CC2 each with their own independent 5.1 kΩ pull-down to GND (two resistors, not one shared – does not rely on

the footprint shorting CC1/CC2 together); VBUS left unconnected – the board is self-powered from +12V, not from the USB bus (PA9 is available for an optional VBUS-sense GPIO, not populated).

- **SWD**: 4-pin header (3V3, SWDIO, SWCLK, GND).

3.3 12 V → 3.3 V regulation

A single linear stage, **LD1117S33TR (SOT-223)**, direct from the board's +12 V rail – not a two-stage USB-VBUS-then-3.3V path like the real Blackpill, since this board's primary (and only) power source is the eFuse's +12 V input. A Schottky (D4, SS14) in series protects against reverse input polarity.

Linear dissipation: $P \approx (12 - 0.4_{D4} - 3.3)\,\mathrm{V} \times I$. At an estimated 50 mA load (MCU + USB PHY idle, no heavy peripheral activity), $P \approx 0.42\,\mathrm{W}$ – acceptable in a SOT-223 with modest copper pour as heatsink. If the 3.3 V load later grows past roughly 120 mA, a linear regulator stops being a reasonable choice and a small buck would be needed instead; not required for the MCU + I^2C telemetry load this project targets.

4 PCB Layout

4.1 Organization

Single sheet (not hierarchical – the board is small enough that visual grouping is clearer and avoids managing hierarchical-sheet pins), organized into labeled blocks: *LTC4215 Controller*, *Protection*, *Power*, *Connectors*, *STM32F4x Microprocessor*, *MCU Support* (reset/boot/crystal/LEDs), *STM32 Decoupling*, *USB-C Connector*, *12V → 3.3V*. Board is 2-layer, same stackup as the original eFuse board (1.51 mm FR4 core, $\varepsilon_r = 4.5$, 1 oz copper).

4.2 Placement priority

In order of sensitivity to trace length/noise: (1) the five 100 nF VDD/VSS decoupling caps, each pegged to its own pin; (2) VCAP1 (2.2 µF); (3) the 25 MHz crystal with a short, symmetric path to PH0/PH1; (4) the VDDA/VREF+ filter (L1 + caps); (5) the NRST cap. Length-insensitive items (BOOT0/BOOT1 pull-downs, LEDs, SWD header) were routed last, wherever space remained.

4.3 USB differential pair

Routed as a matched-length differential pair (D+/D-), with the impedance-controlled section limited to the long connector-to-R16/R17 run only – the short stub tying the USB-C connector's duplicate D+/D- pads together (needed for flip-orientation support) is left at normal track width, since a stub that short does not affect the transmission-line behavior and routing it at full width caused it to visually overlap neighboring pads on the connector footprint (Section 4.4).

Target: 90 Ω differential, per the board's real 2-layer stackup (not a generic assumption). Computed with KiCad's PCB Calculator, Coupled Microstrip mode ($\varepsilon_r = 4.5$, $H = 1.51\,\mathrm{mm}$, $T = 35\,\mu\mathrm{m}$), using $Z_{\mathrm{odd}} = 45\,\Omega$ (half of the 90 Ω differential target) and $Z_{\mathrm{even}} = 100\,\Omega$ (tight coupling – a looser first attempt at $Z_{\mathrm{even}} = 55\,\Omega$ synthesized to an impractical $W = 2.8\,\mathrm{mm}$, $S = 1.97\,\mathrm{mm}$, unroutable on a board this size). Final result: $W = 1.293\,\mathrm{mm}$, $S = 0.297\,\mathrm{mm}$, $Z_{\mathrm{d}} = 90.2\,\Omega$.

4.4 Pitfalls found

Documented here, same reason as the pitfalls section in every other project in this series: so they are not re-discovered from scratch next time.

- **Net name collision (VDD vs. VDD), not a wiring bug.** The LTC4215's VDD (pin 24, tied to +12 V through the sense path) and the STM32's VDD label used the identical text "VDD" on the same flat schematic sheet. Since same-name labels merge into one net regardless of which visual block they sit in, this put 12 V directly on the STM32's power pins. Traced from the PCB netlist (every VDD/VBAT pad of U2 landed on net `"+12V"`) rather than guessed. Fixed by renaming the MCU-side label to `VDD_3V3`.
- **Hidden/stacked pins are real, not a KiCad quirk to work around.** The UFQFPN48 symbol represents VSS pins 35/47/49 as hidden duplicates stacked exactly on top of the single visible VSS (pin 23) – confirmed by reading the symbol's own pin coordinates in the library file. Moving pins around in a locally-edited copy of the symbol is safe only if the hidden duplicates are moved with their visible counterpart; verified after the fact by re-checking coordinates, not assumed safe.
- **A 4-pad SMD crystal footprint is two electrical terminals, not four.** Pads 1/4 (one physical lead, split for solder anchoring) and 2/3 (the other lead) must be tied together with an explicit track – the schematic symbol (`Crystal_Small`, 2 pins) never assigns a net to pads 3/4, so there is no ratsnest guiding the connection; it has to be routed on faith, then verified. Routing pad 3 to pad 4 directly (the two *different* terminals, physically adjacent on the footprint) would short the oscillator – caught before committing to it.
- **PCB-only pad net overrides do not survive "Update PCB from Schematic."** The pad 3/4 net assignment above was first done as a pad-properties override directly in the PCB editor (since the schematic doesn't know about those pads); every re-sync from the schematic silently reverted it, three times over the course of this layout. Root cause understood, not yet fixed permanently (Section 8).
- **A same-net 0 Ω "jumper" resistor still needs its own two pads routed.** Two 0 Ω resistors were used to cross INTVCC and ALERT over an intersecting trace. Because both pads were already the same net per the schematic, KiCad's ratsnest does not treat the resistor body as providing continuity between them – DRC kept reporting them as unconnected until a track was explicitly drawn pad-to-pad across the component itself, which is correct behavior and not a bug.
- **BOOT1 is real on this part, not an F1-series leftover.** Initially assumed the WeAct reference's "BOOT1 (PB2)" pull-down was mislabeled legacy from the F1 series (which has a dedicated BOOT1 pin). Confirmed against RM0383 §2.4: the F401/F410/F411/F412 value-line parts multiplex BOOT1 onto PB2 specifically because they have fewer pins than the rest of the F4 family. The pull-down is required – an undriven PB2 during the boot-mode sample window can make the part start from SRAM instead of entering the USB-DFU bootloader.

5 PDN Analysis

A DC (steady-state) check of the 3.3 V/GND distribution under the STM32's estimated 50 mA load, run with `padne` (github.com/atx/padne), a KiCad-native DC power delivery network solver. DC and

AC answer different questions and neither substitutes for the other: DC voltage drop is pure $I \times R$ through copper and is blind to the decoupling capacitors (an ideal capacitor carries no steady current, so a DC solver cannot tell whether the decoupling bank is adequate); AC impedance-vs-frequency is what actually determines whether the PDN can supply a fast switching transient without the rail sagging. This section covers the DC result; the AC sweep is not yet done (Section 8).

Setup: `!padne` directives placed as plain text in the schematic (not the PCB – `padne` reads sources/loads from the schematic and copper geometry from the board): an ideal 12 V source at J1, an ideal 3.3 V source at U3's output pads, and a 50 mA current load between U2's VDD and VSS pins. The LDO was modeled as an ideal voltage source rather than with `padne`'s dedicated `REGULATOR` directive – the two together (`REGULATOR` + `CURRENT` on the same net) reliably produced a "ground node current is not zero, exactly the specified load current" solver warning that swapping the directives' current-direction parameters did not fix; isolating the problem (VOLTAGE+REGULATOR alone, no load, solved cleanly) showed the fault is in how `padne`'s solver handles that specific element combination, not in this board's setup – a real limitation of what is explicitly an early-stage tool, not a design error. The substitution means this result shows how much the copper itself drops the already-regulated 3.3 V, not the LDO's own regulation behavior (output impedance, current limit), which is a separate question this method does not answer.

Result: Figure 1 – the 3.3 V/GND plane shows at most 148 µV of drop under the 50 mA load, confirming the copper pour is far from being the bottleneck for this current level.

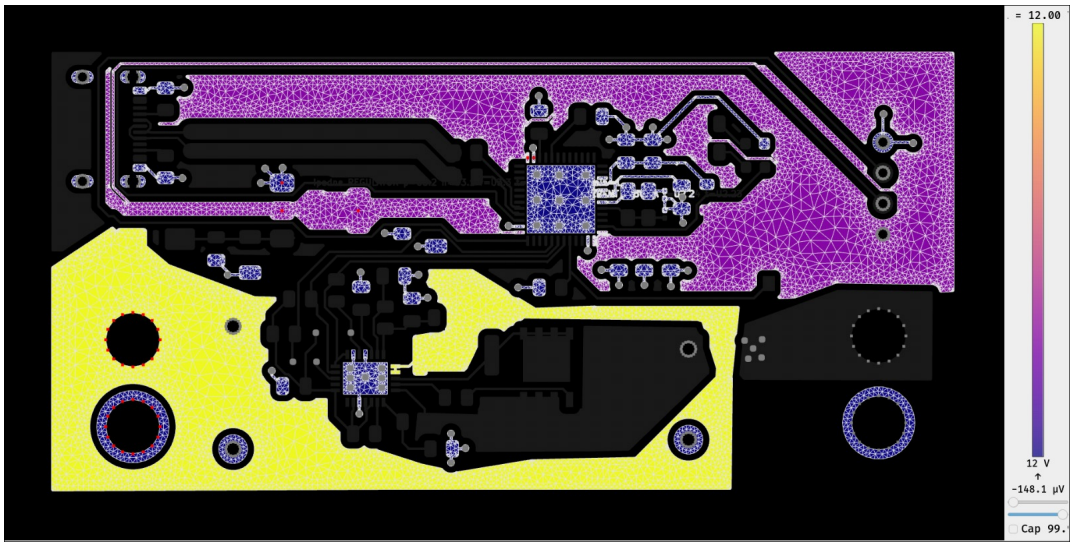


Figure 1: DC voltage distribution (`padne`): yellow/red = 12 V eFuse input, purple = 3.3 V/GND plane under the STM32's 50 mA load. The mesh triangulation is `padne`'s FEM discretization of the copper pours.

AC tooling note: the AC sweep is planned with KiPI (github.com/drewsum/KiPI – not the similarly-named KiPIDA, which is DC-only). Getting it running exposed a real packaging gap: its `requirements.txt` pins `triangle>=20230923`, a meshing dependency with no published wheel past CPython 3.12 and no source distribution on PyPI, so the install silently stalled on newer Python. Worked around by building `triangle` from its GitHub source directly into KiPI's plugin-managed virtual environment.

6 Firmware

An I^{2}C driver for the LTC4215, generated/scaffolded from STM32CubeMX (I2C1 on PB6/PB7, USB_OTG_FS Device-only with VBUS sensing disabled – consistent with Section 3.1 and the self-powered design in Section 8 – and USB_DEVICE configured for CDC/Virtual COM Port) and a hand-written register layer (`Core/Inc/ltc4215.h`, `Core/Src/ltc4215.c`).

Register map: taken directly from the datasheet (Tables 2–8), not re-derived: CONTROL (00h), ALERT (01h), STATUS (02h, read-only), FAULT (03h, write 0 to clear), SENSE (04h, 151 µV LSB), SOURCE (05h, 60.5 mV LSB), ADIN (06h, 4.82 mV LSB – reads ≈0 on this board since ADIN is tied to GND, not used). The main loop polls STATUS/FAULT/SENSE/ SOURCE every 500 ms and writes a line (output current via R5, SOURCE voltage, FET/ power-good/fault flags) out over USB-CDC.

Verification, both without any physical board:

- **Compiles cleanly against the real HAL.** CMake + `arm-none-eabi-gcc`, 0 errors/warnings: 36 020 byte FLASH (6.87% of 512 KB), 9456 byte RAM (7.21% of 128 KB).
- **Register-conversion math checked on the host**, not the target: the SENSE/SOURCE/ADIN raw-byte-to-physical-value functions are factored out of the HAL calls specifically so they can run under plain `gcc` with no STM32 hardware. `code/test/ltc4215_regmap_test.c` feeds synthetic register bytes and checks the result against the datasheet's own LSB values (e.g. raw SENSE= 255 → 38 505 µV → 7701 mA through the 5 mΩ R5) – 11/11 checks pass.

Neither of these is a claim that I^{2}C traffic was ever exchanged with a real LTC4215 – this board was never built. They verify that the code is correct against its two specifications (the HAL's API and the datasheet's register values), which is the honest extent of what is checkable without hardware.

7 Known Limitations

- **LTC4215 is not PMBus-compliant.** It has its own proprietary I^{2}C register map (datasheet Tables 2/3), not the standardized PMBus command set. This project is "digital power monitoring in the spirit of PMBus," not a PMBus device, and is described that way deliberately to avoid overstating the IC's actual capability.
- **Simulation/layout only, not built** – same as every project in this series; no bring-up measurements exist for any of the new circuitry (LDO, USB PHY, I^{2}C timing).

No new thermal analysis was performed for this revision. The LDO's estimated 0.42 W at an assumed 50 mA load is a hand estimate, not simulated; the eFuse power stage's thermal behavior (R5/Q1 at 12 V/15 A) was already characterized in `EFUSE_LTC4215` and is unchanged by this addition.

8 Pending

Pending

- PDN AC impedance-vs-frequency sweep of the 3.3 V rail and its decoupling bank (DC voltage-drop check already done, Section 5), to confirm the capacitor values chosen by convention actually cover the relevant frequency range rather than just assuming "100 nF near every pin is enough." Tooling (KiPI) is installed and loads the board; the probe/sweep setup was not finished this session.
- A permanent fix for the crystal's pad 3/4 net assignment being wiped on every schematic re-sync (either edit the symbol's own pin-to-pad mapping in a local library copy, or simply stop re-running "Update PCB from Schematic" now that the schematic is stable).
- If continuous 15 A operation is ever revisited: same open item as `EFUSE_LTC4215` (heavier copper, metal-strip sense resistor, or airflow) – unaffected by this addition.

9 Conclusions

A minimal STM32F411CEU6 system was added to the LTC4215 eFuse board specifically to exercise the I^2C monitoring interface that the original design had left wired open – SDAI/SDAO tied per the datasheet, the device address reproduced from the datasheet's own worked example rather than invented, and pull-ups referenced to INTVCC after confirming its regulated voltage is close enough to 3.3 V logic levels to skip a level shifter. The MCU side follows the real WeAct Blackpill reference schematic where it was correct and departs from it where it was not: no RTC battery/crystal (not needed here), a single-stage 12 V-direct LDO instead of the Blackpill's two-stage USB-fed supply (this board is never USB-powered), and an independently re-derived, datasheet-confirmed need for the BOOT1 pull-down that was initially (incorrectly) dismissed as an F1-series leftover.

The schematic and PCB are ERC/DRC-clean; the remaining DRC entries are inherent to the USB-C connector footprint itself (NPTH hole clearance to its own mounting holes, and the four intentionally-No-Connect VBUS pads touching each other) rather than unresolved design issues. Several non-obvious layout pitfalls specific to this part – the hidden/stacked VSS pins, the 4-pad crystal footprint's two-terminal reality, same-net labels silently merging across unrelated blocks, and a confirmed (not assumed) BOOT1 requirement on the F411 – are documented in Section 4.4 for the next project that touches this MCU family.

A DC PDN check (Section 5) found the 3.3 V/GND plane drops at most 148 µV under the STM32's load – the copper is not the bottleneck. The I^2C driver (Section 6) compiles cleanly against the real HAL and its register-conversion math is checked against the datasheet's own LSB values on the host, 11/11 – neither claims real hardware was involved, since none was built. The AC PDN impedance sweep remains open, scoped honestly in Section 8 rather than claimed as done.